

CIUDAD REAL, Spain

WMO: 083480

Lat: 38.9892N

Lon: 3.9203W

Elev: 2064

StdP: 13.63

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.8	29.4	17.8	14.6	41.2	21.2	17.1	40.8	23.1	50.1	18.8	47.0	1.4	120	0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.3	100.0	67.6	97.3	67.0	94.9	66.2	70.6	94.1	69.1	92.3	67.8	90.2	6.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.1	93.3	77.4	61.3	87.5	76.0	60.0	83.5	74.9	35.6	94.0	34.4	92.5	33.3	90.3	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	14.5	12.4	21.8	105.1	3.9	2.4	19.0	106.8	16.7	108.1	14.5	109.5	11.6	111.2
			20.7	73.6	3.7	2.9	18.1	75.7	15.9	77.4	13.8	79.1	11.1	81.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.3	43.0	46.2	51.8	56.4	64.6	74.9	80.7	79.2	70.0	61.3	50.4	44.5	(6)
(7)		DBStd	14.17	5.16	5.20	5.56	5.89	6.80	6.39	4.94	4.52	5.82	5.07	5.56	5.41	(7)
(8)		HDD50	654	221	123	47	12	1	0	0	0	2	63	185		(8)
(9)		HDD65	3205	680	526	411	262	91	6	0	18	135	439	637		(9)
(10)		CDD50	4424	5	17	102	205	454	746	952	904	600	353	73	13	(10)
(11)		CDD65	1503	0	0	0	5	79	302	487	439	169	22	0	0	(11)
(12)		CDH74	19355	0	0	13	148	1034	3799	6612	5659	1809	280	1	0	(12)
(13)		CDH80	9901	0	0	0	16	362	1931	3773	3061	702	56	0	0	(13)
(14)	Wind	WSAvg	4.6	4.1	5.3	5.6	5.4	4.9	4.8	4.7	4.8	4.3	3.8	4.0	4.0	(14)
(15)	Precipitation	PrecAvg	16.9	1.8	1.8	1.6	1.8	1.6	1.1	0.3	0.2	1.0	1.9	2.0	2.2	(15)
(16)		PrecMax	27.9	7.0	6.1	5.0	4.7	5.7	5.2	2.8	1.0	4.1	5.4	4.8	8.7	(16)
(17)		PrecMin	9.8	0.0	0.0	0.0	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	4.3	1.5	1.6	1.3	1.1	1.1	1.2	0.6	0.3	0.9	1.5	1.3	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.9	67.2	75.5	82.2	92.5	101.3	103.9	102.1	97.1	85.3	71.2	61.3	(19)
(20)			MCWB	50.4	52.0	56.0	59.4	64.1	67.5	67.7	68.5	66.1	60.7	56.3	51.8	(20)
(21)		2%	DB	57.3	62.9	71.9	78.5	87.8	97.0	100.6	98.7	91.2	80.8	66.4	58.1	(21)
(22)			MCWB	49.0	50.5	53.9	57.9	63.0	66.5	67.8	67.9	65.3	59.6	55.1	52.8	(22)
(23)		5%	DB	54.8	59.4	68.7	74.6	83.8	93.6	98.3	96.3	87.3	76.8	63.0	55.8	(23)
(24)			MCWB	48.9	49.4	52.8	56.2	61.1	65.3	67.6	67.2	64.0	59.5	54.3	51.4	(24)
(25)		10%	DB	52.4	56.3	64.8	70.3	79.9	90.6	95.4	93.6	83.9	73.1	60.2	53.6	(25)
(26)			MCWB	47.5	48.8	51.2	54.3	59.7	64.2	66.2	66.2	63.1	58.9	53.4	49.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	54.9	58.3	61.8	67.3	70.3	72.8	73.4	70.2	64.8	60.8	56.0	(27)
(28)			MCDWB	57.5	61.6	72.6	78.9	87.8	93.9	98.1	94.8	88.1	76.8	65.4	58.1	(28)
(29)		2%	WB	51.8	52.9	55.5	59.0	64.6	68.6	70.5	70.8	67.6	63.0	58.3	53.9	(29)
(30)			MCDWB	55.0	59.4	68.2	74.4	82.4	92.0	95.9	93.4	85.3	73.7	62.9	56.4	(30)
(31)		5%	WB	50.0	51.0	53.9	57.3	62.5	67.2	68.9	69.2	65.9	61.5	56.2	52.0	(31)
(32)			MCDWB	53.3	57.2	65.4	71.7	79.6	89.5	93.8	91.6	82.6	71.2	60.7	55.1	(32)
(33)		10%	WB	48.1	49.4	52.3	55.5	60.7	65.6	67.6	67.7	64.4	60.2	54.3	50.1	(33)
(34)			MCDWB	51.7	55.0	62.5	68.0	77.0	86.3	91.5	89.1	80.2	69.2	58.9	53.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.9	19.1	22.3	21.6	22.5	25.4	26.3	26.5	23.2	20.8	17.6	15.4	(35)
(36)			MCDBR	22.0	26.0	30.3	29.9	30.8	30.2	30.3	30.6	28.8	28.2	22.9	18.0	(36)
(37)			MCWBR	14.8	14.9	14.9	13.1	12.0	10.6	11.1	11.6	11.6	12.5	13.4	12.2	(37)
(38)		5% WB	MCDWB	15.4	19.6	25.9	26.2	27.0	27.1	27.4	27.7	25.0	21.3	16.0	14.1	(38)
(39)			MCWBR	11.7	12.2	13.4	12.3	11.3	10.2	10.6	11.1	10.7	10.2	10.4	10.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.289	0.308	0.343	0.347	0.362	0.366	0.348	0.357	0.350	0.335	0.306	0.291	(40)	
(41)		taud	2.507	2.466	2.379	2.377	2.364	2.375	2.404	2.385	2.418	2.470	2.510	2.516	(41)	
(42)		Ebn at Noon	284	292	290	295	291	288	293	288	284	278	274	274	(42)	
(43)		Edh at Noon	24	29	35	37	39	38	37	36	33	29	24	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	733	1016	1396	1762	2110	2421	2505	2184	1681	1171	816	664	(44)	
(45)		RadStd	105	133	157	145	171	95	62	68	94	98	96	65	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	N/A	N/A	N/A	+1.03	-1.17	-0.99	N/A	N/A	N/A	+124	(47)

Nomenclature: See separate page